Lesson 2.8 Teaching Support Guide

Teach-tomorrow route

Print: current Guided Notes and Practice A for the class; Exit Tickets; optional Language Support and supported homework. Keep this guide and answer keys teacher-facing. Use the full-size model pages in the updated notes.

Phase	Planned minutes	Teaching move
Retrieve	5	Warm-Up items 1–4; note the missing prerequisite before launch.
Launch	7	Name the session target and invite a prediction from the conceptual task.
Model	10	Link a visual model to the worked example; rehearse the mathematical explanation.
Practice	18	Core: A #1, #2, #4, #6 and the error task. Choose remaining items from evidence.
Check	5	Exit Ticket; use the diagnostic page before assigning reteaching.

This is a planned 45-minute route, not a measured classroom duration. For a shorter lesson, keep the model and exit; reduce repeated practice. Form B is parallel practice or reassessment only when its items are unseen.

S1: Finding the Mean · I can find the mean of a data set by sharing the total equally among the values.

S2: Reaching a Target Mean · I can find the value a data set still needs in order to reach a target mean.

Spaced retrieval: revisit one item after 2 lessons and after approximately 1 week. Mix it with the next unit’s prerequisite check; do not announce the operation before students choose a method.

Session 1 · Finding the Mean

I can find the mean of a data set by sharing the total equally among the values. · 6.SP.B.5c

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. 68 ÷ 4 = 17 b. 72 ÷ 6 = 12	Place one container for each group. Count the containers before redistributing the total.
2	410 ÷ 5 = 82 points	Place one container for each group. Count the containers before redistributing the total.
3	C — average	Ask the learner to show why C — average is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Leaves a zero observation out of the count..

Response: Place one container for each group. Count the containers before redistributing the total.

Also check: Reporting the total as the mean (forgetting to divide by the count).

Also check: Leaving out zeros, which divides by too few values.

If calculation alone is wrong, use Session 1 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 1 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

Find and interpret the mean of 0, 3, 6, 7 cans.

Teacher solution: Total 16 divided by 4 = 4 cans per group; keep the zero observation in the count.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: Total 16 cans, four groups; mean 4 cans per group. The zero contributes nothing to the total but is still one group sharing the total. A fair-share model redistributes 6 from the group with 10: 4 to the zero group and 2 to the group with 2.

Misconception response anchor: Leaves a zero observation out of the count.. Use place one container for each group. Count the containers before redistributing the total.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: The mean is ___ per group because ___ objects are shared among ___ groups.

Session 2 · Reaching a Target Mean

I can find the value a data set still needs in order to reach a target mean. · 6.SP.B.5c

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. 54 b. 20	Draw four score slots, including the missing score. Multiply the desired mean by four before subtracting the known total.
2	80 × 5 = 400; 70 + 85 + 78 + 81 = 314; 400 − 314 = 86 points	Draw four score slots, including the missing score. Multiply the desired mean by four before subtracting the known total.
3	A — number of values	Ask the learner to show why A — number of values is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Uses the number of known values rather than the final number of observations..

Response: Draw four score slots, including the missing score. Multiply the desired mean by four before subtracting the known total.

Also check: Multiplying by the old count instead of the new count that includes the missing value.

Also check: Subtracting the target mean from the sum so far instead of building a target total.

If calculation alone is wrong, use Session 2 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 2 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

Scores are 5, 7, and 8. What fourth score gives a mean of 7? Check it.

Teacher solution: Target 4 × 7 = 28; current 20; fourth score 8; (5+7+8+8)/4 = 7.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: For mean 8, target total is 4 × 8 = 32; current total 23; missing score 9. For mean 9, target total 36; missing score 13, impossible with a maximum of 10.

Misconception response anchor: Uses the number of known values rather than the final number of observations.. Use draw four score slots, including the missing score. Multiply the desired mean by four before subtracting the known total.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: There will be ___ scores, so the target total is ___. The missing score is ___, which is/is not possible because ___.